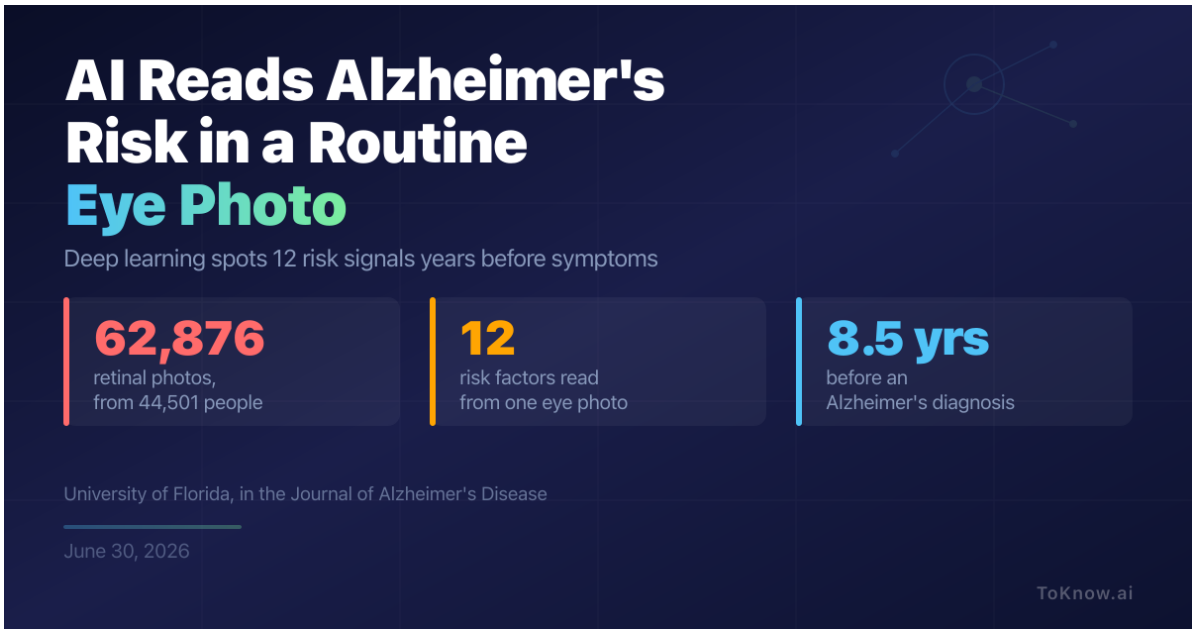
AI Reads Alzheimer's Risk in a Routine Eye Photo

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2026-06-30

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University of Florida researchers trained AI to read ordinary photographs of the retina, the tissue at the back of the eye, for signs of Alzheimer's risk. Using 62,876 eye photos from 44,501 people in the UK Biobank, a large UK health database, two deep-learning models learned to predict 12 factors tied to the disease from a single image, among them blood pressure, smoking, alcohol use, sleeplessness, and blood sugar. The AI nailed some factors and barely beat a coin

flip on others. The telling part: when it marked which pixels mattered, it kept pointing at the optic nerve and retinal blood vessels, the structures a specialist inspects by hand.

Retinal photos are already everywhere. People with diabetes, glaucoma, or cataracts get them routinely, and a checkup for new glasses can capture one. That makes this far cheaper than an MRI, a PET scan, or a spinal tap, the usual ways to probe for Alzheimer's. The retina is the one piece of the central nervous system a camera can see directly, because it grows from the same tissue as the brain. An everyday eye photo doubles as a low-cost sensor of brain health.

The model flags risk, it does not diagnose. The signals turned up in people an average of 8.5 years before an Alzheimer's diagnosis, the window when lifestyle changes or treatment can still help. Like Mayo Clinic's AI reading routine CT scans years early, the shift is from late diagnosis to early warning.

Sources:

- Prediction of Alzheimer's Disease Risk Factors From Retinal Images via Deep Learning (Journal of Alzheimer's Disease)
- Open-access abstract on Europe PMC
- Retinal photographs can help predict Alzheimer's disease risk factors (University of Florida News)
- Retinal Photos and AI Predict Early Alzheimer's Risk (Neuroscience News)

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